and we introduce an initial output token $o \in \mathbb{R}^D$. Let the total sequence length be $n := L + 3$. The initial state (token matrix) is

$$Z^{(0)}(p, x) \;:=\; \begin{bmatrix} p_1 \\ \vdots \\ p_L \\ u(x) \\ w \\ o \end{bmatrix} \in \mathbb{R}^{n \times D}.$$

We denote the i-th token (row) at depth t by $z_i^{(t)} \in \mathbb{R}^D$, i.e., $Z_{i,:}^{(t)} = z_i^{(t)}$.

Row-wise softmax with temperature. For $\tau > 0$, S_τ denotes the row-wise softmax defined by $S_\tau(r) = \mathrm{softmax}(r/\tau)$ for each row vector r, applied row-wise to matrices.

Single-head self-attention. Let $W_Q, W_K, W_V \in \mathbb{R}^{D \times D}$. Define single-head self-attention by

$$A_\tau(Z) \;:=\; S_\tau\left(\frac{(ZW_Q)(ZW_K)^\top}{\sqrt{D}}\right)(ZW_V), \qquad Z \in \mathbb{R}^{n \times D}.$$

(Equivalently, the $\sqrt{D}$ scaling can be absorbed into τ; we keep it explicit for clarity.)

Token-wise FFN. For a hidden width h, define the token-wise two-layer ReLU FFN

$$\phi(z) \;:=\; W_2\,\sigma(W_1 z + b_1) + b_2, \qquad W_1 \in \mathbb{R}^{h \times D},\; W_2 \in \mathbb{R}^{D \times h},$$

where σ is applied elementwise. Extend ϕ row-wise to matrices by

$$(\Phi(Z))_{i,:} \;:=\; \phi(Z_{i,:}) \qquad (i = 1, \ldots, n).$$

Block update and scalar readout. We consider a stack of T blocks. The block update is written as

$$Z^{(t+\frac{1}{2})} \;=\; Z^{(t)} + A_\tau(Z^{(t)}), \qquad Z^{(t+1)} \;=\; Z^{(t+\frac{1}{2})} + \Phi(Z^{(t+\frac{1}{2})}), \qquad t = 0, \ldots, T-1.$$

From the final output token $z_{\mathrm{out}}^{(T)} := Z_{n,:}^{(T)}$, we define the scalar output as

$$F_{\theta^*}(p, x) \;:=\; v_{\mathrm{out}}^\top z_{\mathrm{out}}^{(T)} + b_{\mathrm{out}},$$

where $v_{\mathrm{out}} \in \mathbb{R}^D$ and $b_{\mathrm{out}} \in \mathbb{R}$ are fixed and included in θ^*.

Viewing the prompt as key–value memory and prompt read. For the prompt matrix $P = [p_1; \ldots; p_L] \in \mathbb{R}^{L \times D}$, define slot-wise keys and values by

$$K_P \;:=\; PW_K, \qquad V_P \;:=\; PW_V,$$

where W_K, W_V are the same matrices used inside A_τ. Given a query $q \in \mathbb{R}^D$, define the prompt read operation

$$\mathcal{R}_\tau(q; P) \;:=\; S_\tau\left(\frac{qK_P^\top}{\sqrt{D}}\right) V_P \in \mathbb{R}^D.$$

(As in self-attention, the $\sqrt{D}$ factor can be absorbed into τ; we keep it explicit for consistency with A_τ.)

Remark (simplifications). We do not use LayerNorm, dropout, or multi-head attention. The construction uses only (i) selective readout via attention, (ii) token-wise local computation via the FFN, and (iii) depth-wise composition via residual stacking.

C. Routing Lemma: Prompt-Memory Retrieval

This appendix formalizes the routing primitive used throughout the construction. The key object is the prompt read operation

$$\mathcal{R}_\tau(q;P) \;:=\; S_\tau\left(\frac{qK_P^\top}{\sqrt{D}}\right) V_P \in \mathbb{R}^D,$$

where S_τ is the row-wise softmax with temperature $\tau > 0$ and where, for a prompt matrix $P = [p_1; \ldots; p_L] \in \mathbb{R}^{L\times D}$, we define slot-wise keys and values by

$$K_P := PW_K, \qquad V_P := PW_V,$$

using the same matrices $W_K, W_V \in \mathbb{R}^{D\times D}$ as in $\mathcal{A}_\tau$.

C.1. Margin condition and impurity

Fix a query $q \in \mathbb{R}^D$ and define the score of slot $j \in [L]$ by

$$s_j \;:=\; \langle q, (K_P)_{j,:}\rangle \;=\; (qK_P^\top)_j.$$

Let $j^* \in [L]$ denote the intended slot. We say that (q, P) satisfies a *margin* $\Delta > 0$ for j^* if

$$s_{j^*} \;\geq\; s_j + \Delta \qquad \text{for all } j \neq j^*. \tag{1}$$

Let $\alpha := S_\tau(qK_P^\top) \in \mathbb{R}^{1\times L}$ be the attention weights over prompt slots; then

$$\mathcal{R}_\tau(q;P) \;=\; \sum_{j=1}^{L} \alpha_j (V_P)_{j,:}$$

is a convex combination of prompt values. We refer to $1 - \alpha_{j^*}$ as the *routing impurity*.

C.2. Routing bounds

Lemma C.1 (Softmax routing under a margin)**.** *Fix $\tau > 0$ and let $\alpha = S_\tau(qK_P^\top)$. If the margin condition (1) holds for some $\Delta > 0$, then*

$$\alpha_{j^*} \;\geq\; \frac{1}{1 + (L-1)e^{-\Delta/\tau}}, \tag{2}$$

$$1 - \alpha_{j^*} \;\leq\; \frac{(L-1)e^{-\Delta/\tau}}{1 + (L-1)e^{-\Delta/\tau}} \;\leq\; (L-1)e^{-\Delta/\tau}. \tag{3}$$

Proof. Let $M := \max_{j\neq j^*} s_j$. By (1), $s_{j^*} \geq M + \Delta$. Hence,

$$\sum_{j\neq j^*} e^{s_j/\tau} \;\leq\; (L-1)e^{M/\tau} \;\leq\; (L-1)e^{(s_{j^*}-\Delta)/\tau} \;=\; (L-1)e^{s_{j^*}/\tau}e^{-\Delta/\tau}.$$

Therefore,

$$\alpha_{j^*} = \frac{e^{s_{j^*}/\tau}}{e^{s_{j^*}/\tau} + \sum_{j\neq j^*} e^{s_j/\tau}} \;\geq\; \frac{1}{1 + (L-1)e^{-\Delta/\tau}},$$

and the impurity bound follows by $1 - \alpha_{j^*}$. □

Lemma C.2 (Readout error under bounded values)**.** *Assume the prompt values are uniformly bounded as $\|(V_P)_{j,:}\|_\infty \leq B$ for all $j \in [L]$. Let j^* satisfy (1), and let $\alpha = S_\tau(qK_P^\top)$. Then*

$$\left\|\mathcal{R}_\tau(q;P) - (V_P)_{j^*,:}\right\|_\infty \;\leq\; 2B\,(1 - \alpha_{j^*}) \;\leq\; 2B\,(L-1)e^{-\Delta/\tau}. \tag{4}$$

Proof. Write $v_j := (V_P)_{j,:} \in \mathbb{R}^D$ and $v^* := v_{j^*}$. Then

$$\mathcal{R}_\tau(q;P) - v^* = (\alpha_{j^*} - 1)v^* + \sum_{j \neq j^*} \alpha_j v_j.$$

Taking $\|\cdot\|_\infty$ and using $\|v_j\|_\infty \leq B$ gives

$$\|\mathcal{R}_\tau(q;P) - v^*\|_\infty \leq (1-\alpha_{j^*})\|v^*\|_\infty + \sum_{j \neq j^*} \alpha_j \|v_j\|_\infty \leq 2B(1-\alpha_{j^*}),$$

and the exponential bound follows from (3). □

C.3. A convenient parameter choice

It is often convenient to target a desired impurity level $\rho \in (0,1)$.

Corollary C.3 (Temperature for a desired impurity)**.** *Under the margin condition* (1)*, if*

$$\tau \;\leq\; \frac{\Delta}{\log\left(\frac{L-1}{\rho}\right)},$$

then $1 - \alpha_{j^*} \leq \rho$*. Consequently, under the bounded-value assumption of Lemma C.2,*

$$\left\|\mathcal{R}_\tau(q;P) - (V_P)_{j^*,:}\right\|_\infty \;\leq\; 2B\rho.$$

Proof. By Lemma C.1, it suffices to ensure $(L-1)e^{-\Delta/\tau} \leq \rho$, which is equivalent to the stated condition on τ. The readout bound then follows from Lemma C.2. □

Remark (enforcing the margin). In the construction, prompts store keys such that the query q produced at a given step aligns with the intended key and is separated by a margin Δ from all distractors. The lemmas above isolate the analytic consequence: routing error decays exponentially in Δ/τ.

D. Arithmetic Lemma: Token-wise FFN Computation

This appendix formalizes the arithmetic primitive used in the construction. After routing injects the required information into a token, a *token-wise* two-layer ReLU FFN can approximate the desired local update uniformly on a bounded domain. Concretely, we treat the FFN as a map

$$\phi(z) \;:=\; W_2\,\sigma(W_1 z + b_1) + b_2, \qquad z \in \mathbb{R}^D,$$

and show that, for any continuous update rule on a bounded state set, one can choose (W_1, W_2, b_1, b_2) so that ϕ matches that update to any prescribed uniform accuracy.

D.1. Bounded domain

For $B > 0$, define the compact box

$$\mathcal{B}_B \;:=\; \{z \in \mathbb{R}^D : \|z\|_\infty \leq B\} = [-B, B]^D.$$

All uniform approximation statements below are taken over $\mathcal{B}_B$.

D.2. Piecewise-linear approximation

We first recall that continuous functions on a compact box admit uniform approximation by piecewise-linear maps.

Lemma D.1 (PL approximation on a box)**.** *Let* $f : \mathcal{B}_B \to \mathbb{R}$ *be continuous. For any* $\delta > 0$*, there exists a continuous piecewise-linear function* $\tilde{f} : \mathcal{B}_B \to \mathbb{R}$ *such that*

$$\sup_{z \in \mathcal{B}_B} |f(z) - \tilde{f}(z)| \;\leq\; \delta.$$